

Linux terminal and enable software configuration one by one, as shown in Fig. 6.

```
sudo raspi-config
```

Next, install driver for the E-ink touch display and the Python modules to use the display. To do that, open the Linux terminal and run the following commands:

```
wget http://www.airspayce.com/mikem/
bcm2835/bcm2835-1.68.tar.gz
tar zxvf bcm2835-1.68.tar.gz
cd bcm2835-1.68/
sudo ./configure && sudo make && sudo make
check && sudo make install

sudo apt-get install wiringpi

#For Pi 4, you need to update it:
wget https://project-downloads.drogon.
net/wiringpi-latest.deb
sudo dpkg -i wiringpi-latest.deb
gpio -v
#You will get 2.52 information if you
install it correctly
sudo apt-get update
sudo apt-get install python3-pip
sudo apt-get install python3-pil
sudo apt-get install python3-numpy
sudo pip3 install RPi.GPIO
sudo pip3 install spidev

cd ~
git clone https://github.com/waveshare/
Touch_e-Paper_HAT
```

Creating UI

We need to create the user-interface (UI) for the phone so that it looks beautiful. You can choose any icon and design a custom UI of your choice. For the prototype, round icons were used, as shown in Fig. 7. For UI, we must have the basic design to perform the basic functions. So, here is the list of UI pages for the basic phone functions:

- Main screen
- Dialing screen
- Contacts screen
- Messages
- Incoming call
- Calling screen

For every screen UI we need to

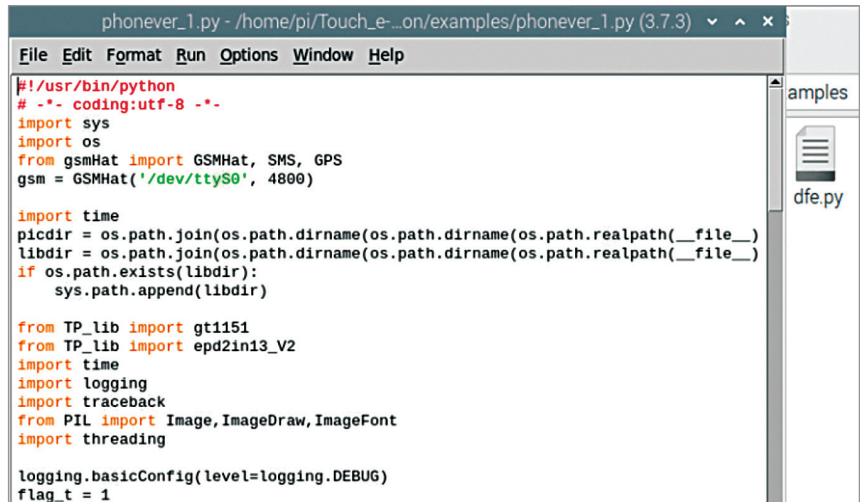


Fig. 9: Python code

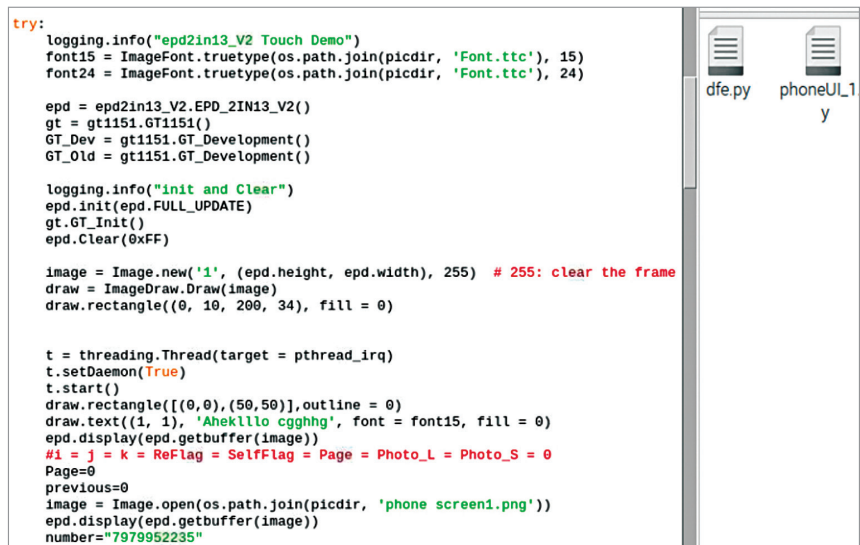


Fig. 10: Home screen

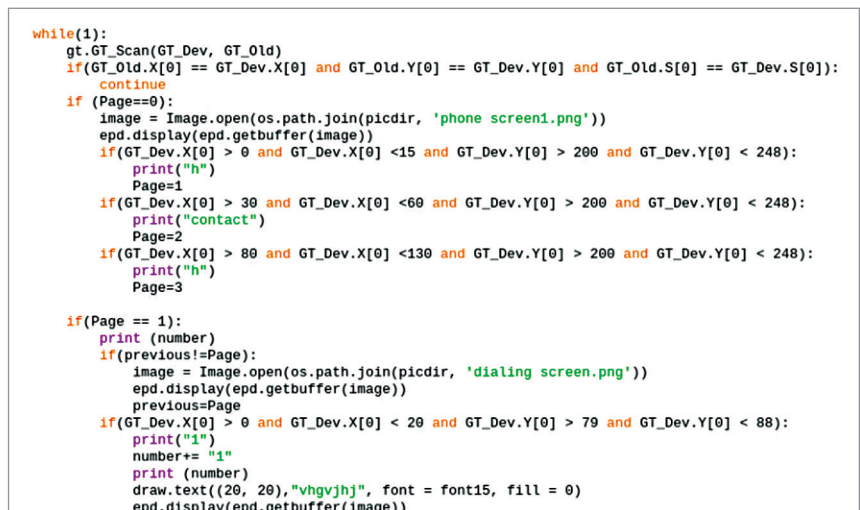


Fig. 11: Call function